WE CLAIM:

1. A virtual sensing system [100] for water quality monitoring (WQM), comprising:

one or more nodal units comprises of two or more sensor nodes S [130], and a central node C [120]; and

a water medium [110],

characterized in that, the central node C [120] is connected to N number of sensor nodes S [130], each sensor node S [130a.....130n] comprises of a transceiver [220], a microcontroller unit [210] and an antenna [230] with two or more physical sensors [240a, 240b, 240c] and two virtual sensors, the central node C [120] comprises of a transceiver [320], a microcontroller unit [310], a display unit [340], a Global System for Mobile Communications (GSM) modem [350], an user device [360], a server [370] and an antenna [330],

wherein the sensor node S [130] is deployed in the water medium [110] to acquire water quality parameters by physical sensors [240a, 240b, 240c], and other water quality parameters [240d, 240e] are virtually sensed using the acquired water quality parameters using physical sensors [240a, 240b, 240c], the sensed signals are converted to respective water quality parameters [240d, 240e] by a virtual sensing module in the microcontroller [210] of the sensor node S [130], the microcontroller [310] of the central node C [120] is provided for controlling and receiving data from the sensor node S [130], and the value of water quality parameters [240d, 240e] is virtually sensed from acquired electrical signals by the physical sensors [240a, 240b, 240c] and one or more others parameters are acquired by using the virtually sensed parameters [240d, 240e], and

wherein the microcontrollers [210, 310] of the nodal units [130, 120] at the sensor node S is embedded with instructions.

2. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the physical sensors [240a, 240b, 240c] acquire water quality parameters such as temperature, pH, conductivity and the like.

3. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the virtual sensing water quality parameters comprises of dissolved oxygen [240d], salinity [240e] and the like.

4. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the microcontroller unit [210, 310] of the nodal units [130, 120] comprises of a memory module and an input/output module and the microcontroller unit [210] also has a virtual sensing module.

5. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the memory module is configured to store the water quality parameters by physical sensors [240a, 240b, 240c] and virtual sensing water quality parameters [240d, 240e], and the input/output module is connected to the physical sensors [240a, 240b, 240c] configured to receive the sensing signals.

6. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the water quality parameters by physical sensors [240a, 240b, 240c] and virtual sensing water quality parameters [240d, 240e] are sent to the central node C [120] from the sensor node S [130] using the transceiver [220, 320] and the antenna [230, 330], received at the central node C [120] and stored in a memory and the central node C [120] communicates the information and uploads the virtual sensing water quality parameters [240d, 240e] to the server [310].

7. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the server [310] is a cloud server.

8. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the microcontroller is configured to use the GSM modem [350] to send water quality parameters by physical sensors [240a, 240b, 240c] and virtual sensing water quality parameters [240d, 240e] information to the user device [360] by the cloud server [310] if online and by the GSM modem [350] if offline.

9. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the user device [360] comprises of one or more of laptop, mobile, smart devices or any other device.

10. A method [400a, 400b] of working of the virtual sensing system [100] for water quality monitoring (WQM), comprising:

initialization [410] of the physical sensors [240a, 240b, 240c] for virtual sensing water quality parameters [240d, 240e] in the water medium [110] and output the sensed signal;

reading [420] the physical sensors [240a, 240b, 240c] and storing the values in the memory by the microcontroller [210];

processing [430] the raw data from each sensor and converts to the virtual sensing water quality parameters values [240d, 240e] by virtual sensing module of the microcontroller [210];

virtual sensing [440] of water quality parameters [240d, 240e];

sending [450] data to the central node C [120];

waiting [460] for predefined time;

receiving [510] data from the sensor node S [130];

storing [520] the data in the memory;

analyzation [530] of the data;

checking [540] whether the parameters are in normal range;

wherein if the parameters are in normal range, the information is communicated [550] to the user and the data is displayed to the user device [360] by the display unit [340],

wherein if the parameters are in abnormal range, the information is communicated [560] to the user by an alarm unit,

waiting [570] for predefined time;

wherein the operations [540] is to determine to alarm the user or pass information in a defined time interval [560], and

displaying the virtual sensing water quality parameters values [240d, 240e] to the user device [360] by the display unit [340],

wherein the process is repeated continuously passing the information to the end-user and uploading the data to the cloud server [370].

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